Field Research Transcript & SME Operational AI Upgrade Benchmark

Simulated SME Field Interview & Operational Quantitative Analysis for Deventrade B.V. and Benelux Teamwear Wholesalers

1. Executive Summary & Operational Baseline

Deventrade B.V. operates as a prominent regional sports apparel and teamwear wholesaler headquartered in Deventer, Overijssel. Managing a central distribution hub of **13,500 square meters** featuring **17 kilometers of pallet racks**, Deventrade distributes sportswear across 14 dedicated e-commerce webshops and supplies over **700 regional B2B sports retailers and club shops**. The primary logistics complexity stems from an active inventory catalog exceeding **18,350 unique SKUs**, structured across a multi-dimensional matrix of Size x Color x Club Badge Printing.

Despite implementing advanced warehouse technologies—including **Centric Locus WMS with CaptureTech voice-picking** and **Akeneo PIM** for catalog enrichment—Deventrade's lean IT team of **1 to 2 staff members** faces severe operational bottlenecks during peak August–September amateur sports league kickoff surges. The core vulnerability is **broken size-run allocations** (running out of core M/L sizes while overstocking XS/XXL), which stalls complete team kit fulfillments and overwhelms customer support reps with manual stock allocation inquiries.

2. Complete Field Research Simulation Transcript

Methodology Note: Conducted under the AIBS & AEL 3-Round Interview Iteration Protocol (Prepare your interview) simulating ground-truth operational parameters for regional teamwear wholesalers.

Part 1: Operational Reality & Daily Logistics Bottlenecks

Patrik Roland Herbert (Product Manager): Jan-Willem, thank you for meeting with our HAN University research team. To ground our study on how regional sports wholesalers evaluate AI, could you walk us through your current operational footprint and where your daily logistics bottlenecks occur?

Jan-Willem van den Berg (SME Operations Lead): Thanks for having me. To understand our operation, you have to look at our variant complexity. We manage over 18,350 unique SKUs across our 13,500 m^2 central warehouse in Deventer. Every jersey or tracksuit is a multi-dimensional matrix of Size x Color x Club Badge. We use voice-picking linked to Centric Locus WMS, which keeps physical picking errors low across our 17 km of pallet racking. But our real pain point isn't picking items off the rack—it's inventory forecasting and broken size runs.

Patrik Roland Herbert (Product Manager): When you say 'broken size runs', what does that look like during your peak ordering cycles?

Jan-Willem van den Berg (SME Operations Lead): Our business is intensely seasonal. In August and September, when Dutch KNVB football and hockey leagues open, we get hit with massive order surges from 700+ regional B2B sports retailers and club webshops. A club orders 30 full team kits. If we run out of Medium or Large jerseys, the entire 30-piece kit order stalls. We end up with plenty of XS or XXL stock sitting on pallet racks, but we cannot fulfill the complete team set. That creates sudden stockouts in core

sizes, inflates dead stock in non-core sizes, and swamps our back-office team with manual stock inquiries across our 14 web portals.

David Malena (Developer): How large is your internal IT and data team to handle these ERP inventory adjustments?

Jan-Willem van den Berg (SME Operations Lead): We operate like most regional Dutch wholesalers in Gelderland and Overijssel: we have 1 to 2 IT staff members. Their entire day is consumed maintaining core ERP uptime, keeping server connections stable, and troubleshooting dealer shop syncs. We do not have data engineers, Python developers, or in-house machine learning specialists. Any AI tool we adopt must run reliably without requiring constant custom code maintenance.

Part 2: Addressing Research Proposal Sub-Questions

Patrik Roland Herbert (Product Manager): That context directly addresses our research proposal. Let's run through five specific areas our proposal covers to evaluate where an AI upgrade is viable.

Sub-Question 1: Vendor Landscape & SME Tool Availability

Patrik Roland Herbert (Product Manager): Desk research shows enterprise supply-chain AI platforms built for multinational retail chains. From an SME perspective, who actually builds tools that fit a mid-sized wholesaler?

Jan-Willem van den Berg (SME Operations Lead): Enterprise platforms built for massive retail chains are completely out of reach—their implementation fees exceed our annual IT budget, and they assume dedicated data engineering teams. For mid-sized SMEs, we see three vendor categories: (1) Specialized European SaaS add-ons that plug into standard Dutch ERPs like AFAS, Exact Globe, or SAP Business One; (2) Third-party AI wrappers that present fancy dashboards but carry high lock-in risks; and (3) Pragmatic workflow automation CLI/API tools that clean historical sales data and calculate seasonal reorder triggers without touching our core WMS.

Sub-Question 2: Failure Modes & Model Calibration

Jan Dragon (Tester): In our desk research, we studied common failure modes in AI demand forecasting. What specific risks would make you reject an AI recommendation?

Jan-Willem van den Berg (SME Operations Lead): The biggest failure point is uncalibrated model drift during seasonal spikes. Generic AI models look at linear sales history. They see low jersey sales in June and July and predict low demand for August. But in sports teamwear, June/July is off-season, and August is league kickoff. If an AI model blindly projects off-season trends into August, it causes catastrophic stockouts. Second, silent failures. If an automated script fails or outputs a hallucinated size-run order without alerting us, we lose weeks of lead time. We require an explicit confidence floor—an 85% confidence threshold. If model confidence drops below 85%, the system must flag a warning and default to historical ERP reorder averages.

Sub-Question 3: Legacy ERP Data Cleanliness & Volume

David Malena (Developer): How far back does your historical sales log go in your ERP, and how clean is that data for machine learning model ingestion?

Jan-Willem van den Berg (SME Operations Lead): We have 5 to 7 years of raw ERP sales logs. However, raw data is messy. Historical size-run entries are sometimes logged under inconsistent unit codes, club badge custom printing notes are recorded in free-text fields, and obsolete SKUs clutter the database. If you feed raw ERP tables directly into an AI model, you get garbage in, garbage out. The first

step in any technical upgrade isn't building a predictive model—it's deploying an automated local data-cleaning script that standardizes legacy size-run metadata before forecasting occurs.

Sub-Question 4: Data Privacy, PII Scrubbing & EU Compliance

Daniel Kelemen (Deployer): We designed our platform architecture around strict data privacy guardrails. What are your non-negotiable boundaries regarding uploading operational data to external APIs?

Jan-Willem van den Berg (SME Operations Lead): We have strict non-negotiable boundaries under GDPR and EU data residency laws: Zero Raw PII Uploads. Client names, B2B dealer contact details, and club representative phone numbers must never leave our local network or be uploaded to public US cloud models. Our purchase margins, cost prices, and supplier discount structures are strictly confidential IP. All raw interview notes, customer emails, and ERP sales logs must undergo automated local PII and financial anonymization on our local machine before any text snippet is sent to an external model API. External models should only process anonymized SKU sequences.

Sub-Question 5: Practical Deliverables & ROI KPI Alignment

Patrik Roland Herbert (Product Manager): What structure makes our final research report genuinely actionable for an SME operations manager?

Jan-Willem van den Berg (SME Operations Lead): Do not give me a 50-page theoretical treatise on neural networks. Structure your output across Three Maturity Tiers and anchor everything in concrete Dutch SME KPIs: Tier 1 (Off-the-Shelf) prompt guardrails for dealer support; Tier 2 (Workflow Automation) local CLI scripts for automated ERP data scrubbing and PIM enrichment; and Tier 3 (Predictive Operations) demand-planning layers balancing core size runs. Most importantly, conclude with a standalone 1-Page Quick-Start Action Sheet that my IT lead and I can review in 10 minutes. Link every recommendation directly to reducing cut-to-ship cycle times, lowering fabric scrap and overstock rates, and maintaining a >99% order accuracy SLA.

3. Quantitative Operational Metrics & AI Upgrade Benchmark Matrix

The table below synthesizes the empirical operational benchmarks of Deventrade B.V. against the pragmatic 3-Tier AI implementation framework established in the AIBS & AEL curriculum:

Operational Dimension	Baseline SME Reality (Deventrade Benchmark)	Target AI Upgrade & Maturity Tier
IT & Engineering Capacity	1 to 2 generalist IT staff members managing ERP uptime and network servers; zero data engineers or ML specialists.	Tier 2 Local CLI Scripting: Automated data cleaning scripts with manual fallbacks (zero ML team required).
Warehouse & SKU Scale	13,500 m^2 hub, 17 km pallet racks, 18,350+ SKUs across Size x Color x Club Badge variants, 14 web portals.	Tier 2 PIM Catalog Enrichment: Automated daily PIM metadata generation across e-commerce portals via Akeneo API.
Seasonal Demand Bottleneck	Aug/Sept league kickoff surges cause broken size runs (Medium/Large stockouts while XS/XXL remain overstocked).	Tier 3 Demand-Planning Threshold: Automated reorder validation with an explicit 85% confidence floor to balance size runs.

Operational Dimension	Baseline SME Reality (Deventrade Benchmark)	Target AI Upgrade & Maturity Tier
Data Privacy & Governance	Strict GDPR & EU AI Act obligations; confidential B2B dealer margins and raw ERP customer contact details.	Tier 2 Local Anonymization Pipeline: On-premise script stripping PII and margin metrics prior to external API ingestion.
Core Operational KPIs	Order processing bottlenecks, dead stock accumulation, and manual customer service allocation inquiries from 700+ B2B dealers.	Measurable Target Metrics: Reduced Cut-to-Ship Cycle Time, 12-15% Fabric Scrap/Overstock Reduction, and >99% Order Accuracy SLA.

4. Source Legend & Primary Reference Register

This research simulation and operational analysis are grounded in the following primary industry case studies, corporate announcements, and academic course materials:

1. **Deventrade B.V. Corporate Logistics Profile (2026):** Centralized warehouse infrastructure (13,500 m^2, 17 km pallet racks, 18,350+ active SKUs, 14 e-commerce portals, 700+ sports dealer network). Deventer, Overijssel, Netherlands.
2. **Centric Locus WMS Case Study — Deventrade (2025):** 'Deventrade kiest voor de betrouwbaarheid van Locus WMS'. Details warehouse management system migration, real-time inventory tracking, and voice-guided order picking across high-density pallet racks.
3. **Akeneo PIM Customer Story — Deventrade (2024):** Automated Product Information Management case study detailing daily catalog enrichment and multi-channel publication across 14 webshops and 1,000 variant groups.
4. **Adwise E-Commerce Growth Case — Deventrade (2025):** 'Deventrade versnelt e-commerce groei met Hyvä Commerce'. Benchmarks e-commerce frontend acceleration and B2B portal performance.
5. **CaptureTech Voice Picking Implementation (2024):** 'Deventrade kan USP nog beter benutten dankzij voice-picking'. Examines warehouse picking accuracy and 17 km rack navigation efficiency.
6. **Indeed Netherlands Employee Reviews — Deventrade B.V. (2025):** Field feedback highlighting seasonal warehouse workload spikes and manual inventory re-allocation during August-September sports league openings.
7. **Regional Benelux Competitors Benchmark (HAN Minor AIBS/AEL 2026):** Comparative analysis of direct sports teamwear wholesalers: Robey Sportswear (Rotterdam), Derbystar / SELECT Sport Benelux (Linden), NOOR Sportswear (Eindhoven), Sportconfex (Assen), and Tricorp Workwear (Rijen).
8. **Draghi Report on European Competitiveness — EU SME Diffusion (2024):** Address by Mario Draghi on European economic competitiveness, highlighting the productivity gap in European mid-sized firms due to slow technology diffusion.